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Lecture Notes

22MA201 – ENGINEERING MATHEMATICS II

UNIT I & FIRST ORDER LINEAR DIFFERENTIAL EQUATIONS

(Unit I & LP5)

1. FALLING OBJECTS

1.1 Falling Objects Problems:

Consider a vertically falling body of mass m that is being influenced only by gravity g and an air resistance that is propositional to the velocity of the body. Assume that both gravity and mass remain constant and choose the downward direction as the positive direction.

1.2 Newton's Second Law of Motion:

Newton's second law of motion states that "Force is equal to the rate of change of momentum". For a constant mass, force equals mass times acceleration.

$$F = ma$$

$$F = m\frac{dv}{dt} \qquad ---------(1)$$

where F is the net force on the body and v is the velocity of the body at time t.

1.3 Forces acting on the falling objects:

There are two forces acting on the body:

1. the force due to gravity given by the weight w of the body, $w = mg$.

2. the force due to air resistance is called the **drag force**. It is always in the direction opposite of velocity. If the force due to air resistance is **proportional to the velocity** and in the opposite direction then `drag force' = $-kv$, where $k \geq 0$ is called the drag coefficient.

1.4 Note:

1. The value of the gravity is $g = 9.8\, m/s^2 = 32\, ft/s^2$.
2. The limiting velocity of the object is $v_l = \dfrac{mg}{k}$.

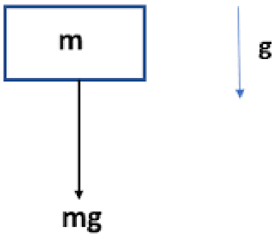
1.5 Differential equations of falling object:

1. Free Fall (No air resistance):

In this case, the net force acting on the body is $F = mg$, then the equation of motion is (see equation (1))

$$m\frac{dv}{dt} = mg$$

$$\boxed{\frac{dv}{dt} = g}$$



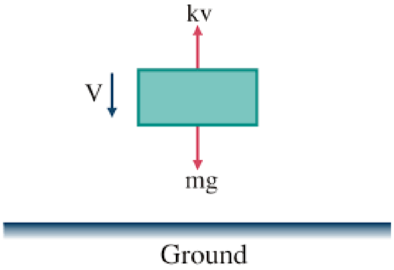
2. Falling with air resistance:

i. If the object is moving **downward**, then the net force acting on the body is $F = mg - kv$, then the equation of motion is (see equation (1))

$$m\frac{dv}{dt} = mg - kv$$

$$\frac{dv}{dt} = g - \frac{k}{m}v$$

$$\boxed{\frac{dv}{dt} + \frac{k}{m}v = g}$$



ii. If the object is moving **upward**, then the net force acting on the body is $F = -mg - kv$, then the equation of motion is (see equation (1))

$$m\frac{dv}{dt} = -mg - kv$$

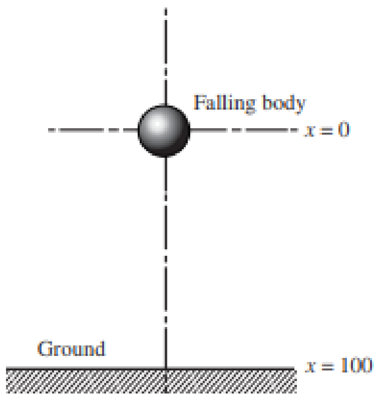
$$\frac{dv}{dt} = -g - \frac{k}{m}v$$

$$\boxed{\frac{dv}{dt} + \frac{k}{m}v = -g}$$

1.6 Problems:

1. A body of mass 5 *lb* is dropped from a height of 100 ft with zero velocity. Assuming no air resistance, find
   (a) an expression for the velocity of the body at any time t,
   (b) an expression for the position of the body at any time t, and
   (c) the time required to reach the ground.

Solution



a) Since there is no air resistance,

$$\frac{dv}{dt} = g \quad \Rightarrow \quad \frac{dv}{dt} = 32 \quad \Rightarrow \quad v = 32t + c$$

When t = 0, v = 0 (initially the body has zero velocity); hence 0 = 32(0) + c, or c = 0. Thus, the velocity of the body at any time t is

$$v(t) = 32t \, .$$

b) Recall that velocity is the time rate of change of displacement, designated here by x.

$$\frac{dx}{dt} = v \quad \Rightarrow \quad \frac{dx}{dt} = 32t$$

$$\Rightarrow \quad dx = 32t\, dt$$

$$\Rightarrow \quad x = 16t^2 + c_1$$

But at t = 0, x = 0 (see Fig). Thus, $0 = (16)(0)^2 + c_1$, or $c_1 = 0$, we have $x = 16t^2$. Thus, the position of the body at any time t is

$$x(t) = 16t^2.$$

c) We require t when x = 100.

$$x(t) = 16t^2 \quad \Rightarrow \quad 100 = 16t^2 \quad \Rightarrow \quad t = 2.5 \text{ sec}$$

2. A steel ball weighing 2 lb is dropped from a height of 3000 ft with no velocity. As it falls, the ball encounters air resistance numerically equal to v/8 (in pounds), where v denotes the velocity of the ball (in feet per second). Find
   (a) the velocity for the ball at any time
   (b) the limiting velocity for the ball and
   (c) the time required for the ball to hit the ground.

Solution

Here w = 2 lb and k = 1/8. Assuming gravity g is 32 ft / $\sec^2$, we have from the formula w = mg that is 2 = m(32) or that the mass of the ball is m = 1/16 .

a) The equation of motion is

$$\frac{dv}{dt} + \frac{k}{m} v = g$$

$$\frac{dv}{dt} + 2v = 32$$

which has as its solution

$$v(t) = ce^{-2t} + 16 .$$

At t = 0, we are given that v = 0. Substituting these values in the above equation, we obtain

$$0 = ce^{-2(0)} + 16 \;\Rightarrow\; c = -16$$

Hence, the velocity for the ball at any time

$$v(t) = -16e^{-2t} + 16 \qquad ----------(1)$$

b) From (1), we see that as $t \to \infty,\; v \to 16$ so the limiting velocity is 16 ft/sec2 .

c) Since v = dx/dt, equation (1) can be rewritten as

$$\frac{dx}{dt} = -16e^{-2t} + 16 \qquad \Rightarrow \qquad dx = (-16e^{-2t} + 16)dt$$

$$\Rightarrow \qquad x = 8e^{-2t} + 16t + c_1 \qquad -----(2)$$

where c_1 denotes a constant of integration. At t = 0, x = 0. Substituting these values into (2), we obtain

$$0 = 8e^{-2(0)} + 16(0) + c_1$$

$$c_1 = -8$$

The position of the ball at any time t is

$$x(t) = 8e^{-2t} + 16t - 8$$

The ball hits the ground when x(t) = 3000. Substituting this value into (3), we have

$$3000 = 8e^{-2t} + 16t - 8$$

$$3008 = 8e^{-2t} + 16t$$

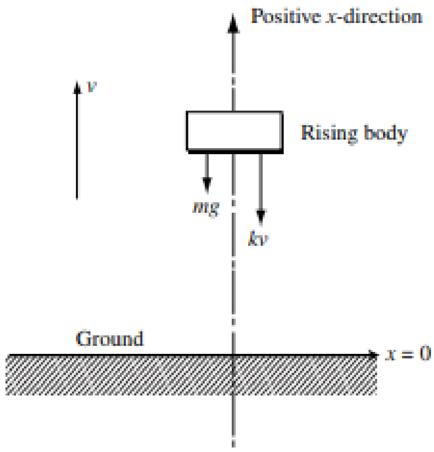
$$376 = e^{-2t} + 2t$$

We note that for any large value of t, the negative exponential term will be negligible. A good approximation is obtained by setting 2t = 376 or t = 188 sec. For this value of t, the exponential is essentially zero.

3. A ball with mass 0.15 kg is thrown upward with initial velocity 20 m/s from the roof of a building 30 m high. There is a force due to air resistance of magnitude v/30 directed opposite to the velocity, where the velocity v is measured in m/s.

   (a) Find an expression for the velocity of the ball at any time t
   (b) Find the maximum height above the ground that the ball reaches.

Solution :



(a) The differential equation of for the given scenario is

$$\frac{dv}{dt} + \frac{k}{m}v = -g$$

Here, $m = 0.15, \quad k = \frac{1}{30}, \quad v(0) = 20\ m/s, \quad x(0) = 30\,m$

Therefore,

$$\frac{dv}{dt}+\frac{1}{4.5}v=-9.8$$

$$4.5\frac{dv}{dt}+v=-44.1$$

$$4.5\frac{dv}{dt}=-(v+44.1)$$

$$\frac{dv}{(v+44.1)}=-4.5\,dt$$

$$\int\frac{dv}{(v+44.1)}=-\int 4.5\,dt$$

$$\log(v+44.1)=\frac{-t}{4.5}+c$$

$$v+44.1=ce^{\frac{-t}{4.5}}$$

$$v=ce^{\frac{-t}{4.5}}-44.1$$

Since v(0) = 20, we must have c = 64.1. Therefore the velocity of the ball at any time t is

$$v(t)=64.1e^{\frac{-t}{4.5}}-44.1$$

(b) The ball reaches its maximum height when v = 0. Solving for t gives

$$0 = 64.1e^{\frac{-t}{4.5}} - 44.1$$

$$64.1e^{\frac{-t}{4.5}} = 44.1$$

$$e^{\frac{-t}{4.5}} = \frac{44.1}{64.1}$$

$$t = 1.683\,s$$

The height of the ball is given by

$$\frac{dx}{dt} = 64.1e^{\frac{-t}{4.5}} - 44.1$$

$$x = \int \left(64.1e^{\frac{-t}{4.5}} - 44.1 \right) dt$$

$$x = -288.45e^{\frac{-t}{4.5}} - 44.1t + c$$

$$x(t) = -288.45e^{\frac{-t}{4.5}} - 44.1t + 318.45 \qquad (\text{since } x(0) = 30)$$

When t = 1.863 s,

x = 45.783 m.

Therefore, the maximum height that the ball reaches is x = 45.783 m.

1.7 Practice Problems:

1. A body of mass 3 slugs is dropped from a height of 500 ft in a with zero velocity. Assuming no air resistance,

(a) find an expression for the velocity of the body at any time t
(b) find an expression for the position of the body at any time t
(c) determine the time required for the body to hit the ground
(d) how long would it take if instead the mass of the body was 10 slugs?

Answers:

$(a)\ v = 32t$ $\qquad (b)\ x = 16t$ $\qquad (c)\ t = 5.59\,\text{sec}$ $\qquad (d)\ t = 5.59\,\text{sec}$

2. A body weighing 64 lb is dropped from a height of 100 ft with an initial velocity of 10 ft/sec. Assume that the air resistance is proportional to the velocity of the body. If the limiting velocity is known to be 128 ft/sec.

   (a) construct the differential equation for given scenario.
   (b) find an expression for the velocity of the body at any time t and
   (c) find an expression for the position of the body at any time t.

   Answers:

   $$(a)\ \frac{dv}{dt}+\frac{1}{4}v=32$$

   $$(b)\ v(t)=-118e^{\frac{-v}{2}}+128$$

   $$(c)\ x(t)=472e^{\frac{-v}{2}}+128t-472$$

<u>Discussion question</u>

1. A body of mass 3 *lb* is dropped from a height of 500 ft with zero velocity. Assuming no air resistance, find

(a) an expression for the velocity of the body at any time t,

(b) an expression for the position of the body at any time t, and